



CSE623 Machine Learning: Theory and Practice

Topic:

Machine Learning Based Risk Parity Portfolio Construction for Financial Markets

Weekly Report 6

Submitted to: Prof. Mehul Raval

Submitted on 28th March 2026

Team Name: RiskNeutral

Team Members:

Bhavya Sharma	AU24110006	<i>Master of Science in Quantitative Finance</i>
Ritik Rakesh	AU24110010	<i>Master of Science in Quantitative Finance</i>
Smit Kalathiya	AU24110013	<i>Master of Science in Quantitative Finance</i>
Shree S	AU24110008	<i>Master of Science in Quantitative Finance</i>
Vivek Maskar	AU24110011	<i>Master of Science in Quantitative Finance</i>

1. Overview of Project:

Risk parity has emerged as a widely adopted portfolio construction framework that allocates capital by equalizing risk contributions across assets rather than maximizing expected returns. Popularized in institutional asset management as an “all weather” strategy, risk parity portfolios are designed to deliver stable performance across different market regimes. However, the effectiveness of risk parity critically depends on accurate estimation of asset volatilities and correlations.

Recent advances in machine learning and statistical learning theory provide tools for denoising and dynamically estimating covariance structures. Techniques such as shrinkage estimators, random matrix theory based eigenvalue filtering, and hierarchical clustering provide more robust representations of dependency structures (tail risks) for a multi-asset portfolio.

This project builds a risk parity portfolio using machine learning techniques. Using daily US financial assets data from 2015–2025, we construct long-short and long-only portfolios using PCA-based risk parity and Hierarchical Risk Parity (HRP). We evaluate their out-of-sample performance from 2020 to 2025 and compare them against the S&P 500 benchmark.

2. Overview of Week 6:

Tasks accomplished this week:

- Executed the HRP rolling back-test with all three covariance estimators: raw sample covariance, Ledoit–Wolf constant-correlation shrinkage, and Marčenko–Pastur denoised covariance. Each variant was evaluated over the same 2020–2025 out-of-sample period.
- Conducted a head-to-head comparison of the three HRP variants, reporting total return, annualized return, annualized volatility, Sharpe ratio, Sortino ratio, maximum drawdown, and Calmar ratio for each estimator.
- Analyzed portfolio weight distributions for the MP-denoised HRP configuration, examining weight concentration, maximum/minimum weights over time, and the number of active assets at each rebalance.
- Computed rolling 252-day correlation between the HRP (MP) strategy and the S&P 500 benchmark to assess whether HRP achieves genuine factor diversification.

3. Work Completed:

HRP Back-Test: All Three Covariance Estimators

- The rolling HRP back-test was executed three times, varying only the covariance estimator while keeping all other parameters identical (4-year training window, 21-day holding, Ward linkage, $1.25\times$ explosion guard). This isolates the effect of covariance estimation quality on HRP allocation.
- The raw sample covariance estimator provides the baseline: it uses no denoising or shrinkage, making it vulnerable to estimation noise in high-dimensional settings ($N=474$ assets). The Ledoit–Wolf estimator shrinks toward a constant-correlation target, reducing off-diagonal noise. The Marčenko–Pastur estimator replaces noise eigenvalues with their mean, directly targeting the eigenvalue distortion that degrades portfolio construction.

Head-to-Head Performance Comparison

- All three HRP variants were evaluated using the same performance metrics framework from Week 4. The comparison reveals how covariance estimation quality propagates through the HRP pipeline: better covariance estimates lead to more accurate correlation distances, which in turn produce more meaningful cluster structures and ultimately better-diversified allocations.
- The results demonstrate that HRP is more robust to covariance estimation quality than the PCA+GMV pipeline. While the PCA approach produced near-zero Sharpe ratios with certain estimators, HRP maintains positive risk-adjusted returns across all three configurations, confirming López de Prado's (2016) finding that HRP's recursive bisection is inherently more stable than methods requiring matrix inversion.

Weight Distribution Analysis

- The weight distribution for the MP-denoised HRP was analyzed at each rebalancing date. Unlike the PCA pipeline, which produced extreme weights reaching $\pm 9,900\%$ before regularization, HRP weights are bounded in $[0, 1]$ by construction. The maximum single-asset weight was tracked over time and remained moderate, confirming that HRP naturally diversifies across the asset universe.
- A histogram of weights at the final rebalance date shows a right-skewed distribution, with most assets receiving small allocations and a few assets receiving larger weights corresponding to low-variance, low-correlation positions. This is consistent with the inverse-variance weighting within each cluster.

Equity Curve Comparison

- The three equity curves (Raw, Ledoit–Wolf, Marčenko–Pastur) were plotted on the same axis for visual comparison. All three show positive performance over 2020–2025, with drawdowns concentrated in the COVID-19 period (early 2020). The MP-denoised variant shows the smoothest equity curve, consistent with the cleaner covariance input reducing allocation noise.

Benchmark Comparison: S&P 500 Rolling Correlation

- A 252-day rolling correlation between HRP (MP) returns and S&P 500 returns was computed. The analysis reveals extended periods of low or negative correlation, indicating that HRP achieves genuine diversification across non-equity risk factors. This contrasts with a naive equity-only portfolio, which would show near-perfect correlation with the benchmark.
- Periods of higher correlation correspond to market stress events where cross-asset correlations spike, a well-documented phenomenon. Post-crisis, the HRP correlation with the benchmark returns to low levels, supporting the “all-weather” hypothesis.

4. Challenges Encountered:

- The Ledoit–Wolf estimator is computationally expensive for the full 474-asset universe within each rolling window, as it requires computing a sum of N outer products. This made the Ledoit–Wolf back-test significantly slower than the other two configurations.
- Interpreting performance differences between covariance estimators is complicated by the fact that the back-test period (2020–2025) includes the COVID-19 crash, the subsequent recovery, and the 2022 rate-hiking cycle. Different estimators may perform better in different regimes, making it difficult to declare a single winner.

- The weight explosion guard, while effective for the PCA pipeline, may be less necessary for HRP since weights are naturally bounded. However, we retained it for consistency and to prevent sudden allocation shifts during market stress.

5. Plan for Week 7:

- Conduct a direct comparison between the PCA+GMV pipeline (from Week 4) and the HRP pipeline across all performance metrics to assess which approach better serves the risk parity objective.
- Document the complete mathematical framework for HRP, covering the three phases (clustering, seriation, recursive bisection) with full derivations and a worked numeric example.
- Finalize the codebase, clean documentation, and upload to GitHub.
- Complete the IEEE conference-style final report consolidating all methodology, results, and analysis from Weeks 1–7.

References:

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